

Electromagnetic Field Theory

* Scalar dot product:

• If $A = (x_1, y_1, z_1)$ & $B = (x_2, y_2, z_2)$

then,

$$\vec{A} \cdot \vec{B} = a_x b_x + a_y b_y + a_z b_z$$

→ here,

$$\vec{a}_x \cdot \vec{a}_x = \vec{a}_y \cdot \vec{a}_y = \vec{a}_z \cdot \vec{a}_z = 1$$

$$\vec{a}_x \cdot \vec{a}_y = \vec{a}_y \cdot \vec{a}_x = \vec{a}_x \cdot \vec{a}_z = \vec{a}_z \cdot \vec{a}_x = 0$$

* Cross product:

• If $A = (x_1, y_1, z_1)$ & $B = (x_2, y_2, z_2)$

then,

$$\vec{A} \times \vec{B} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

→ here,

$$\vec{a}_x \times \vec{a}_x = \vec{a}_y \times \vec{a}_y = \vec{a}_z \times \vec{a}_z = 0$$

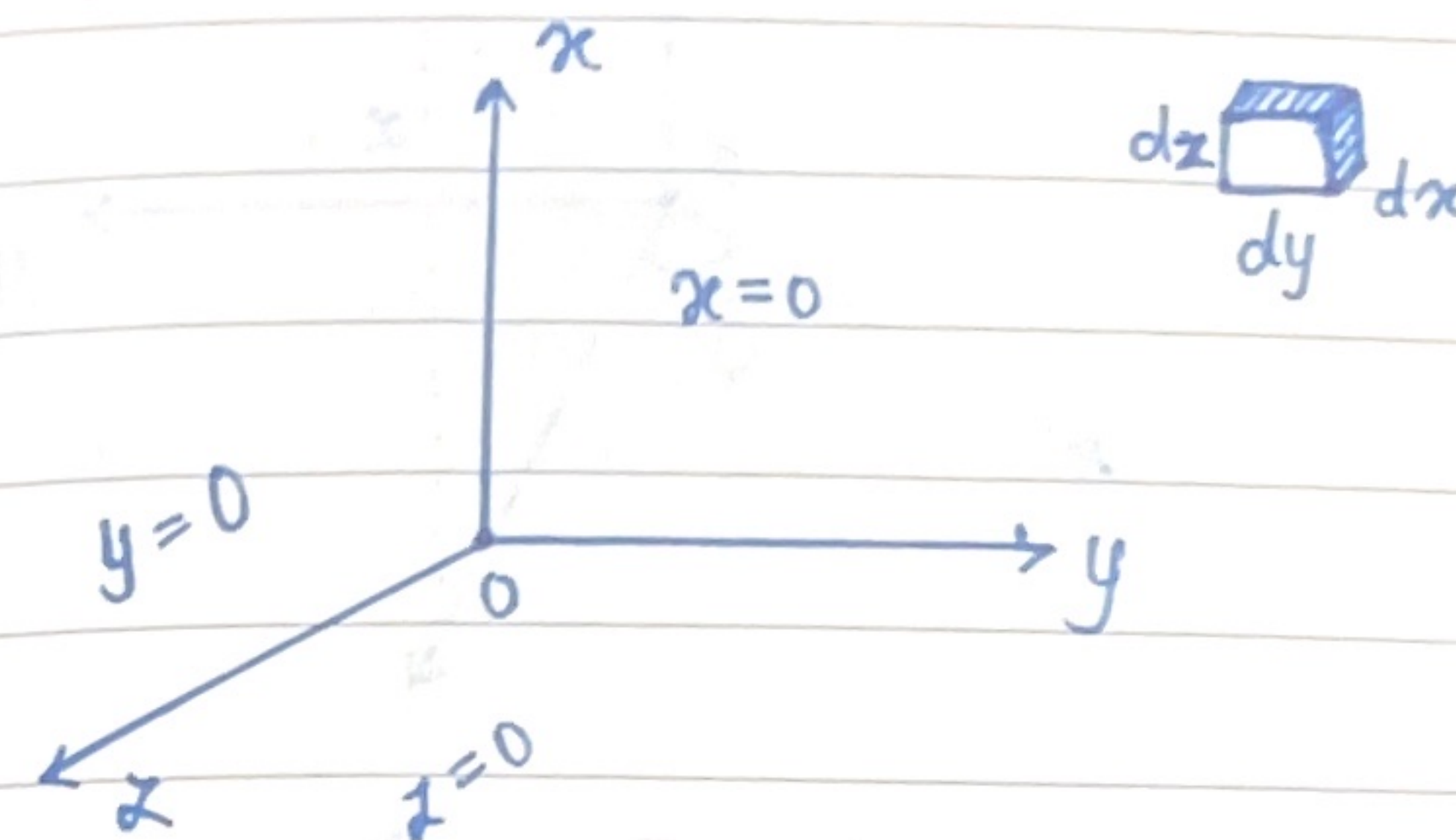
$$\vec{a}_x \times \vec{a}_y = \vec{a}_z, \quad \vec{a}_y \times \vec{a}_x = -\vec{a}_z$$

$$\vec{a}_y \times \vec{a}_z = \vec{a}_x, \quad \vec{a}_z \times \vec{a}_y = -\vec{a}_x$$

$$\vec{a}_z \times \vec{a}_x = \vec{a}_y, \quad \vec{a}_x \times \vec{a}_z = -\vec{a}_y$$

* Co-ordinate Systems

1) Cartesian System (x, y, z):



→ unit vectors: $\hat{a}_x, \hat{a}_y, \hat{a}_z$

→ differential length: (dx, dy, dz)

∴ diff. vector length:

$$dl = dx \cdot \vec{a}_x + dy \cdot \vec{a}_y + dz \cdot \vec{a}_z$$

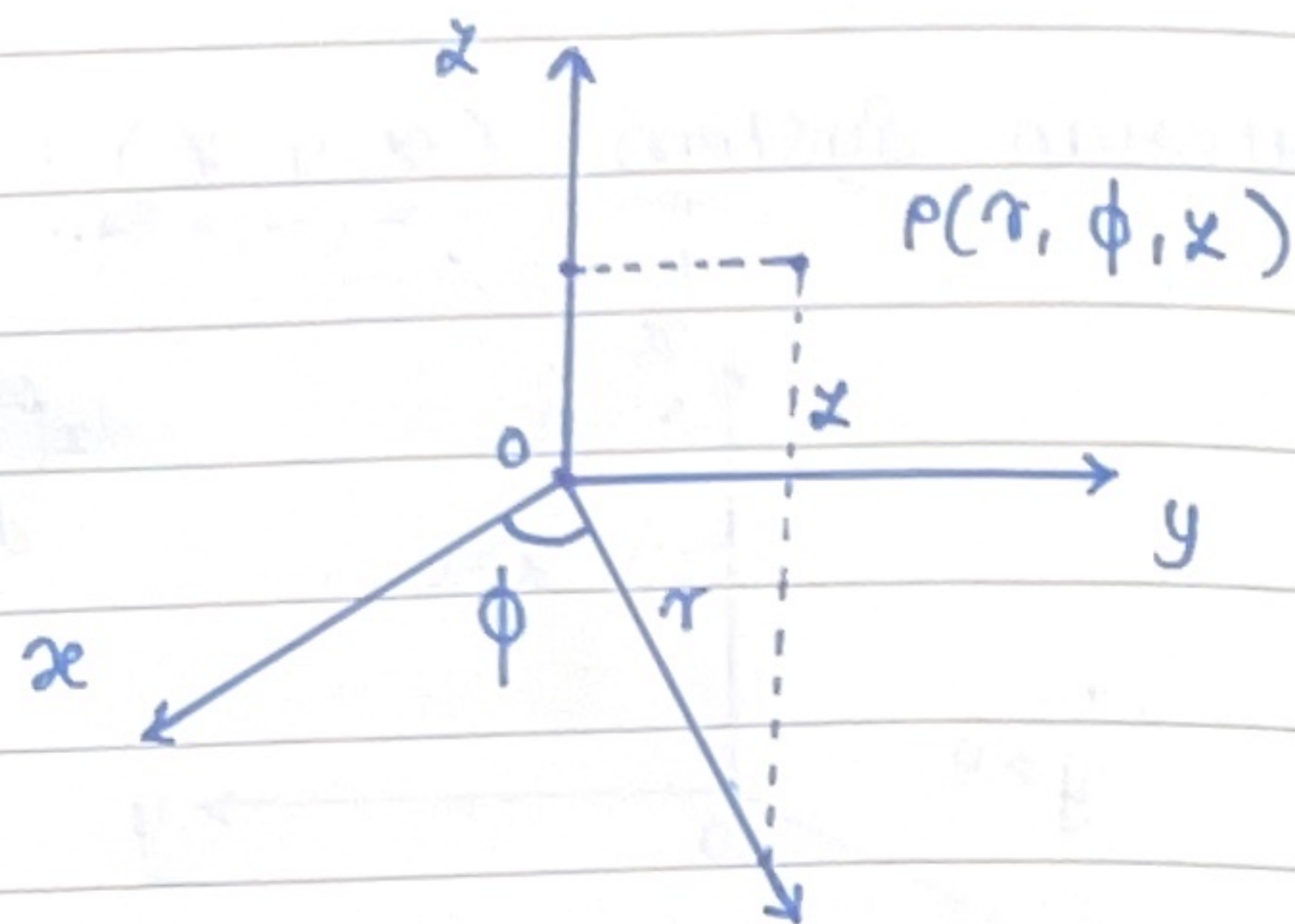
→ diff. area:

$$dS_1 = dy \, dz \quad (x: \text{const.})$$

$$dS_2 = dx \, dz \quad (y: \text{const.})$$

$$dS_3 = dx \, dy \quad (z: \text{const.})$$

→ diff. volume: $dv = dx \cdot dy \cdot dz$

2) Cylindrical System (r, ϕ, z):

→ Unit vectors : $\hat{a}_r, \hat{a}_\phi, \hat{a}_z$

→ diff. length : $(dr, r \cdot d\phi, dz)$

∴ diff. vector length :

$$dl = dr \cdot \bar{a}_r + r \cdot d\phi \cdot \bar{a}_\phi + dz \cdot \bar{a}_z$$

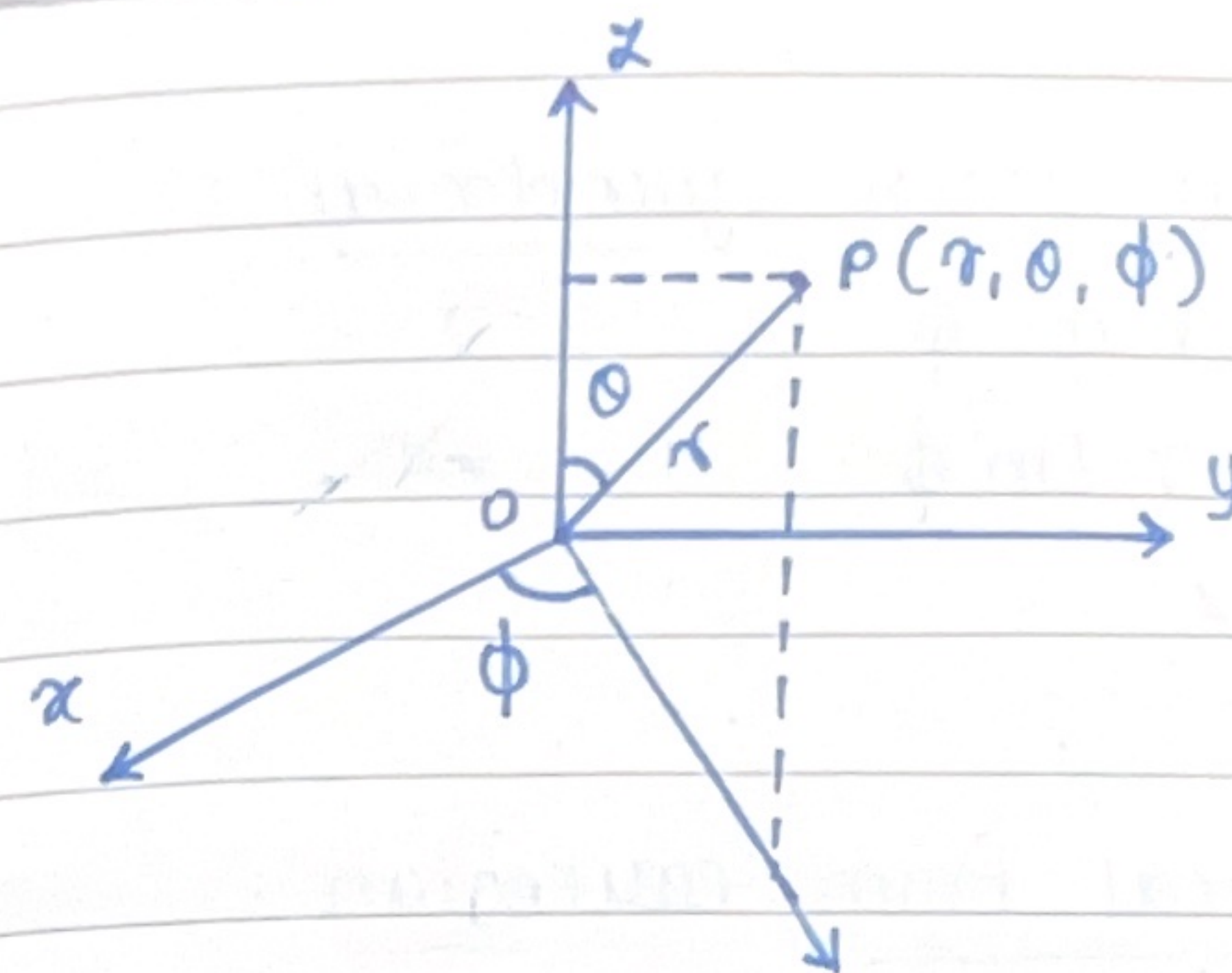
→ diff. area :

$$dS_1 = r \cdot d\phi \cdot dz \cdot \bar{a}_r \quad (r: \text{const.})$$

$$dS_2 = dr \cdot dz \cdot \bar{a}_\phi \quad (\phi: \text{const.})$$

$$dS_3 = r \cdot dr \cdot d\phi \cdot \bar{a}_z \quad (z: \text{const.})$$

→ diff. volume : $dv = r \cdot dr \cdot d\phi \cdot dz$

3) Spherical System (r, θ, ϕ):

→ Unit vectors : $\hat{a}_r, \hat{a}_\theta, \hat{a}_\phi$

→ diff. length : $(dr, r \cdot d\theta, r \cdot \sin \theta \cdot d\phi)$

∴ diff. vector length

$$dl = dr \cdot \bar{a}_r + r \cdot d\theta \cdot \bar{a}_\theta + r \sin \theta \cdot d\phi \cdot \bar{a}_\phi$$

→ diff. area :

$$dS_1 = r^2 \sin \theta \cdot d\theta \cdot d\phi \cdot \bar{a}_r \quad (r: \text{const.})$$

$$dS_2 = r \sin \theta \cdot dr \cdot d\phi \cdot \bar{a}_\theta \quad (\theta: \text{const.})$$

$$dS_3 = r \cdot dr \cdot d\theta \cdot \bar{a}_\phi \quad (\phi: \text{const.})$$

→ diff. volume : $dv = r^2 \sin \theta \cdot dr \cdot d\theta \cdot d\phi$

* Conversion formula :1) Cartesian from cylindrical :

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

2) Cylindrical from Cartesian :

$$r = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}(y/x)$$

$$z = z$$

3) Cartesian from Spherical :

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

4) Spherical from Cartesian :

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1} \left[\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right]$$

$$\phi = \tan^{-1}(y/x)$$

5) Cylindrical from Spherical :

$$r = r_s \sin \phi$$

$$\theta = \theta$$

$$z = r_s \cos \phi$$

6) Spherical from cylindrical :

$$r_s = \sqrt{r^2 + z^2}$$

$$\phi = \tan^{-1} \left[\frac{r}{z} \right]$$

$$\theta = \theta$$

Electrostatics* Coulomb's Law :

- It states that betn. 2 pt. charges there is a force of attraction or repulsion depending on nature of charges, the F is given mathematically ...

$$F = \frac{k q_1 q_2}{r^2} \text{ (N)} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \text{ (N)}$$

here, $k = 9(10^9)$, r : (destn. - source)

$$\epsilon_0 = 8.85(10^{-12}) \text{ F/m}$$

→ Vector Form of Coulomb's Law:

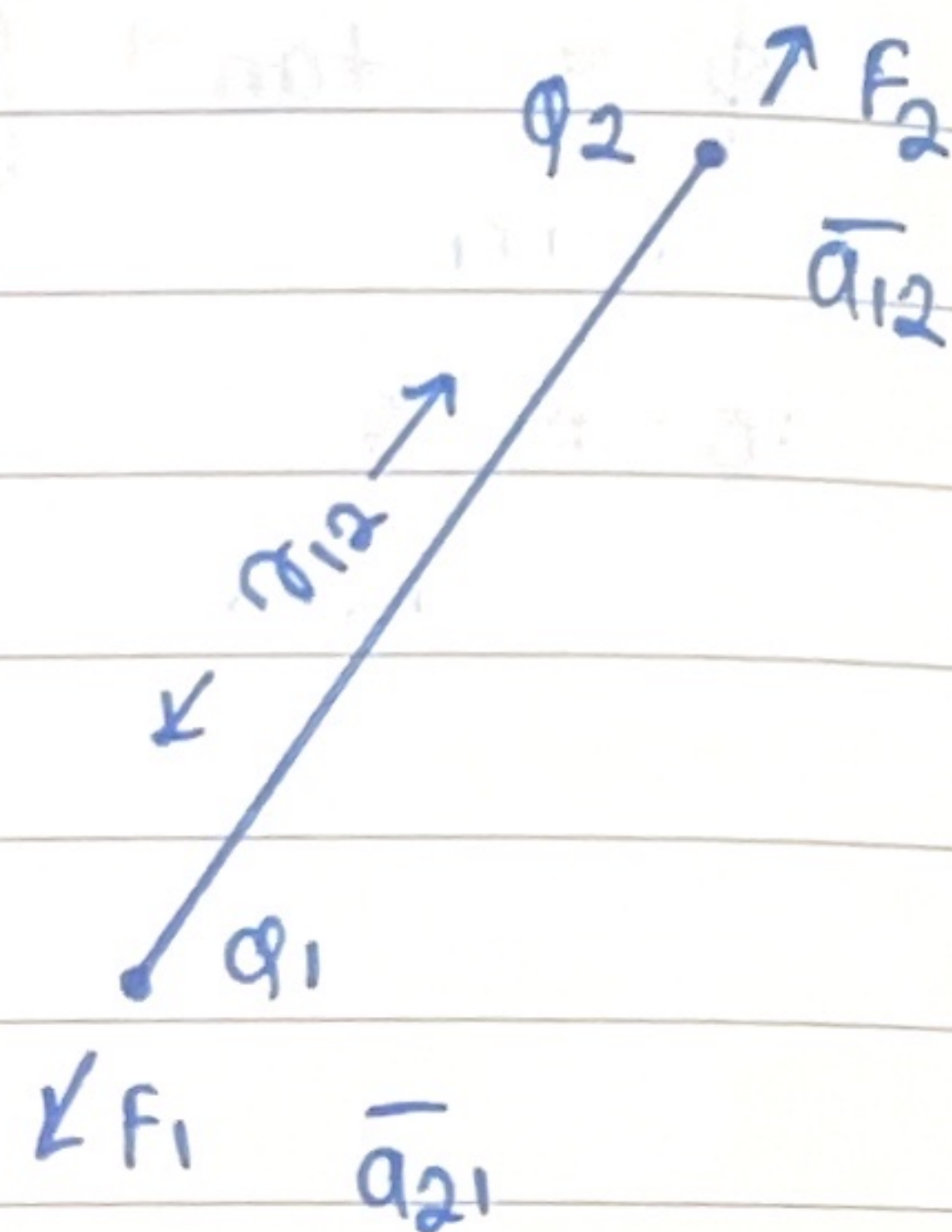
Force on q_1, q_2 : F_1, F_2 respectively

$$F_2 = \frac{q_1 q_2}{4\pi\epsilon_0 r_{12}^2} \bar{a}_{12}$$

F_1 is in opp. dirn. of F_2

$$\therefore F_1 = -F_2$$

$$\text{ie. } F_1 = \frac{q_1 q_2}{4\pi\epsilon_0 r_{12}^2} \bar{a}_{21}$$

→ Example :

Given: $q_1 = 2 \text{ mC}$ at $P_1(-3, 4, -4)$

$q_2 = 5 \text{ mC}$ at $P_2(2, 4, -1)$

To find: F_1 & F_2

Solution :

$$\hat{a}_{12} = \frac{\bar{r}_{12}}{|\bar{r}_{12}|} = \frac{5\bar{a}_x - 3\bar{a}_y + 3\bar{a}_z}{\sqrt{43}}$$

$$\begin{aligned} \therefore F_2 &= \frac{k q_1 q_2}{r_{12}^2} \hat{a}_{12} \\ &= \frac{(9)(10^9)(2)(5)(10^{-6})}{(\sqrt{43})^2} \left[\frac{5\bar{a}_x - 3\bar{a}_y + 3\bar{a}_z}{\sqrt{43}} \right] \\ &= (3.188)(10^2)(5\bar{a}_x - 3\bar{a}_y + 3\bar{a}_z) \text{ N} \end{aligned}$$

• and hence,

$$\begin{aligned} F_1 &= -F_2 \\ &= -318.8 [5\bar{a}_x - 3\bar{a}_y + 3\bar{a}_z] \text{ N} \end{aligned}$$

* Principle of Superposn.

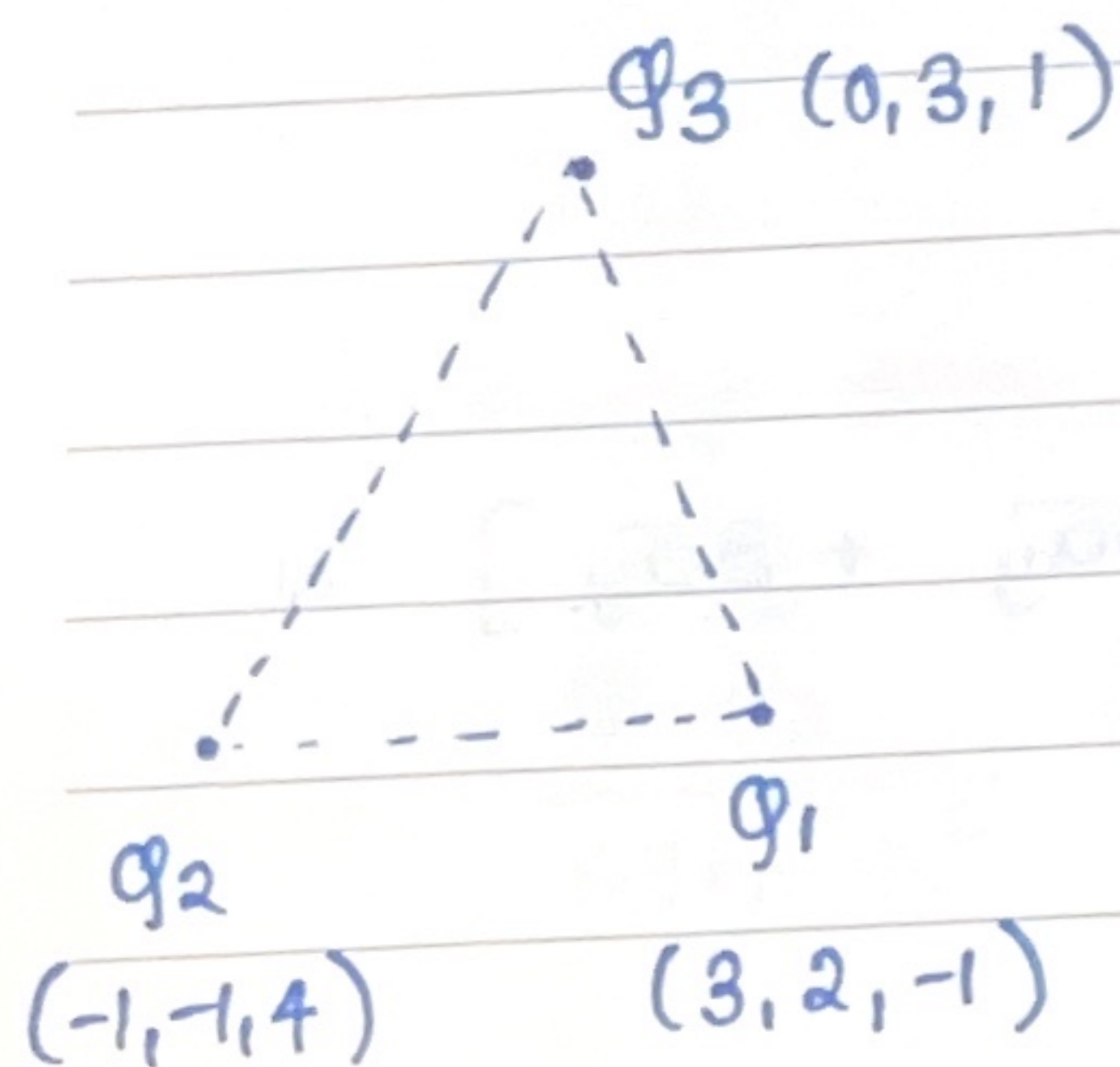
- If the F. on the particular charge to be calculated due to other charges then principle of superposn. can be used.
- In which find the F. on that charge due to other q's separately & total F. is the vector addition of all forces.

→ Example:

Given: $q_1 = 1 \text{ mC}$ at $(3, 2, -1)$

$q_2 = (-2) \text{ mC}$ at $(-1, -1, 4)$

To find: Electric F. on 10 nC at $(0, 3, 1)$

Solution:

$$\begin{aligned} \hat{a}_{13} &= \frac{\mathbf{R}_{13}}{|\mathbf{R}_{13}|} \\ &= \frac{-3\bar{a}_x + \bar{a}_y + 2\bar{a}_z}{\sqrt{14}} \end{aligned}$$

$$\begin{aligned} \hat{a}_{23} &= \frac{\mathbf{R}_{23}}{|\mathbf{R}_{23}|} \\ &= \frac{\bar{a}_x + 4\bar{a}_y - 3\bar{a}_z}{\sqrt{26}} \end{aligned}$$

$$\begin{aligned} \therefore F_{13} &= \frac{(9)(10^9)(10^{-3})(10)(10^{-9})}{14} \left[\frac{-3\bar{a}_x + \bar{a}_y + 2\bar{a}_z}{\sqrt{14}} \right] \\ &= 1.41 (-3\bar{a}_x + \bar{a}_y + 2\bar{a}_z) \text{ mN} \end{aligned}$$

$$\begin{aligned} \therefore F_{23} &= \frac{(9)(10^9)(-2)(10^{-9})(10)(10^{-9})}{26} \left[\frac{\bar{a}_x + 4\bar{a}_y - 3\bar{a}_z}{\sqrt{26}} \right] \\ &= -1.35 (\bar{a}_x + 4\bar{a}_y - 3\bar{a}_z) \text{ mN} \end{aligned}$$

• Thus,

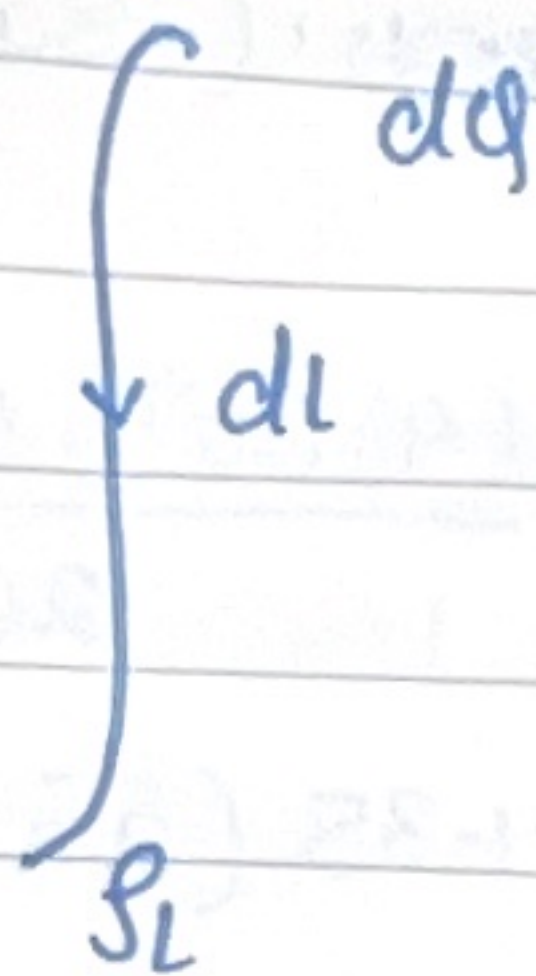
$$\begin{aligned} F_{\text{tot}} &= F_{13} + F_{23} \\ &= (-6.52\bar{a}_x - 3.72\bar{a}_y + 7.52\bar{a}_z) \text{ mN} \end{aligned}$$

NOTES

Date

* line charge:

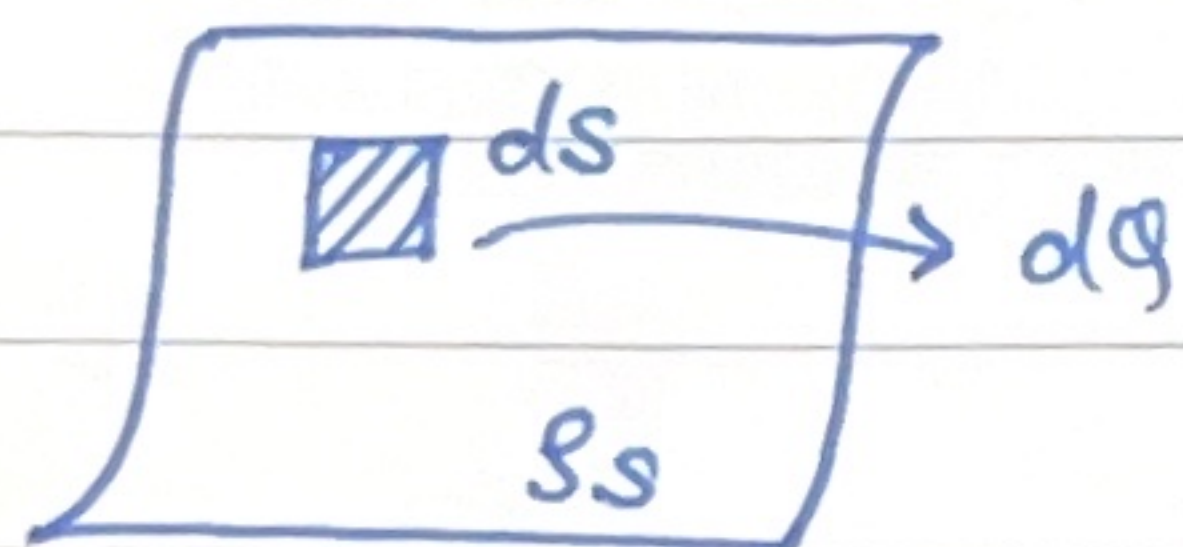
- For the small intersection of line called as 'dl'
- When charge is uniformly distributed over the line, the line charge is described as total charge spread over length of the line



ie. $\rho_L = \frac{dq}{dl}$, here, $dq = \rho_L \cdot dl$

$$\therefore q = \int_L \rho_L \cdot dl$$

* Surface charge:



- When charge is uniformly distributed over a surface

$$\rho_s = \frac{dq}{ds} \Rightarrow dq = \rho_s \cdot ds$$

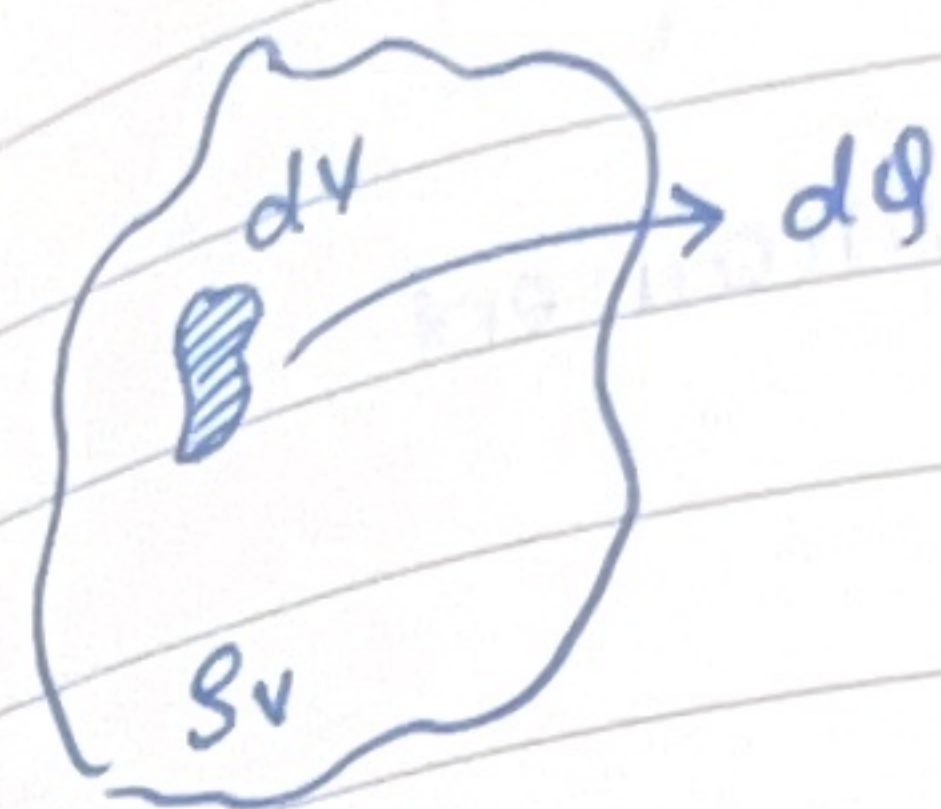
$$\therefore q = \int_S \rho_s \cdot ds = \iint \rho_s \cdot ds$$

NOTES

Date

* volume charge:

- charge is uniformly filled into the volume



$$\rho_v = \frac{dq}{dv}$$

$$\therefore dq = \rho_v \cdot dv$$

$$\therefore q = \int_V \rho_v \cdot dv$$

$$\therefore q = \iiint \rho_v \cdot dv$$

* Differential force:

$$d\vec{F} = \frac{q_1 \cdot dq}{4\pi\epsilon_0 \cdot R^2} \hat{a}_R$$

→ Example:

Given: $\rho_v = (x + 2y + 3z) \text{ C/m}^3$
in region of $0 \leq x, y, z \leq 1 \text{ mm}$
 $\rho_v = 0$, outside

To find: total charge contained inside cube

Solution:

$$Q = \int_V \rho_v \cdot dv$$

$$= \int_0^1 \int_0^1 \int_0^1 (x + 2y + 3z) dx dy dz$$

$$= \int_0^1 \int_0^1 \left[\frac{x^2}{2} + 2yx + 3zx \right]_0^1 dy dz$$

$$= \int_0^1 \int_0^1 \left[\frac{1}{2} + 2y + 3z \right] dy dz$$

$$= \int_0^1 \left[\frac{y}{2} + y^2 + 3zy \right]_0^1 dz$$

$$\therefore Q = \int_0^1 \left[\frac{3}{2} + 3z \right] dz = 3(10^{-12}) = 3 \text{ pF} \rightarrow [\because d \text{ in mm}]$$

→ Example:

$$\rho_v = \frac{3\pi \cos^2 \theta \cdot \cos^2 \phi}{2r^2 (r^2 + 1)} ; \text{ universe}$$

Solution:

• If it is universe: $r \rightarrow 0 \text{ to } \infty$
 $\theta \rightarrow 0 \text{ to } \pi$

$$\phi \rightarrow 0 \text{ to } \pi/2$$

• $\therefore$ these are spherical co-ordinates

$$\therefore dv = r^2 \sin \theta dr d\theta d\phi$$

$$\therefore Q = \int_0^\infty \int_0^\pi \int_0^{\pi/2} \rho_v (r^2 \sin \theta \cdot dr d\theta d\phi)$$

* Electric field:

- Field surrounding electric charge particles where F can be felt is known as Electric field (E)

$$\therefore E = \frac{F_T}{q_T} \quad (E: \text{Electric field intensity})$$

q_T : +ve test charge

- We know,

$$F_T = \frac{q_1 q_T}{4\pi\epsilon_0 R^2} \hat{a}_R$$

$$\therefore E = \frac{F_T}{q_T}$$

$$\therefore E = \frac{q_1}{4\pi\epsilon_0 R^2} \hat{a}_R = k \frac{q_1}{R^2} \hat{a}_R$$

(V/m) or (N/C)

→ Dot product tables:

1) Cartesian and cylindrical

	a_r	a_ϕ	a_z
a_x	$\cos \phi$	$-\sin \phi$	0
a_y	$\sin \phi$	$\cos \phi$	0
a_z	0	0	1

2) Cartesian and Spherical

	a_r	a_θ	a_ϕ
a_x	$\sin \theta \cdot \cos \phi$	$\cos \theta \cdot \cos \phi$	$-\sin \phi$
a_y	$\sin \theta \cdot \sin \phi$	$\cos \theta \cdot \sin \phi$	$\cos \phi$
a_z	$\cos \theta$	$-\sin \theta$	0

→ Example:

A charge q is located at origin in free space produces a field for which

$$E_z = 1 \text{ kV/m at pt. } (-2, 1, -1)$$

Find: a) q b) $\vec{E}$ at $M(1, 6, 5)$

c) $\vec{E}$ at M in cylindrical

d) $\vec{E}$ at M in spherical

→ Solution:

A) Now,

$$\hat{a}_r = \frac{\vec{R}}{|\vec{R}|} = \frac{-2\vec{a}_x + \vec{a}_y - \vec{a}_z}{\sqrt{6}}$$

Now,

$$\vec{E} = \frac{q(9)(10^9)}{6} \left[\frac{-2\vec{a}_x + \vec{a}_y - \vec{a}_z}{\sqrt{6}} \right]$$

$$\therefore E_x = \frac{q(9)(10^9)}{6\sqrt{6}} [-\vec{a}_x]$$

$$\therefore (10^3) = -\frac{q(9)(10^9)}{6\sqrt{6}}$$

$$\therefore q = -1.632 \mu\text{C}$$

B) $\vec{E}$ at $M(1,6,5)$

ie origin to M



$$\begin{aligned} \therefore \vec{R} &= (1-0)\vec{a}_x + (6-0)\vec{a}_y + (5-0)\vec{a}_z \\ &= \vec{a}_x + 6\vec{a}_y + 5\vec{a}_z \end{aligned}$$

$$\therefore \hat{a}_r = \frac{\vec{a}_x + 6\vec{a}_y + 5\vec{a}_z}{\sqrt{62}}$$

$$\therefore \vec{E} = -\frac{(1.632)(10^{-6})(9)(10^9)}{62} \left[\frac{\vec{a}_x + 6\vec{a}_y + 5\vec{a}_z}{\sqrt{62}} \right]$$

$$= (-30\vec{a}_x - 180.5\vec{a}_y - 150\vec{a}_z) \text{ V/m}$$

let it be E_1 ↑

$$c) M(1,6,5) \equiv M(x,y,z)$$

$$\therefore r = \sqrt{x^2 + y^2 + z^2} = \sqrt{37}$$

$$\therefore \phi = \tan^{-1}(y/x) = 80.54^\circ$$

$$\therefore z = 5$$

• Now

$$\rightarrow E_{r1} = E_1 \hat{a}_r$$

$$= (-30\vec{a}_x - 180.5\vec{a}_y - 150\vec{a}_z) \hat{a}_r$$

$$= [-30\vec{a}_x \hat{a}_r - 180.5\vec{a}_y \hat{a}_r - 150\vec{a}_z \hat{a}_r]$$

$$= -30(\cos \phi) - 180.5(\sin \phi) - 0$$

$$= 183.48 \text{ V/m}$$

$$\rightarrow E_{\phi_1} = E_1 \cdot \bar{a}_{\phi}$$

$$= [-30 \bar{a}_x - 180.5 \bar{a}_y - 150 \bar{a}_z] \bar{a}_{\phi}$$

$$= -30 (\bar{a}_x \cdot \bar{a}_{\phi}) - 180.5 (\bar{a}_y \cdot \bar{a}_{\phi}) - 150 (\bar{a}_z \cdot \bar{a}_{\phi})$$

$$= -30 (-\sin \phi) - 180.5 (\cos \phi) - 150 (1)$$

$$= 0$$

$$\rightarrow E_{x_1} = E_1 \bar{a}_x$$

$$= [-30 \bar{a}_x - 180.5 \bar{a}_y - 150 \bar{a}_z] \bar{a}_x$$

$$= -30 (\bar{a}_x \cdot \bar{a}_x) - 180.5 (\bar{a}_y \cdot \bar{a}_x) - 150 (\bar{a}_z \cdot \bar{a}_x)$$

$$= -150$$

$\therefore$ cylindrical co-ordinates

$$\therefore E_2 = 183.48 \bar{a}_x - 150 \bar{a}_y$$

D) Similarly, for spherical co-ordinates calculate:

r, θ and ϕ

then,

$$\bar{E}_{r_1} = \bar{E}_1 \bar{a}_r$$

$$\bar{E}_{\theta_1} = \bar{E}_1 \bar{a}_{\theta}$$

$$E_{\phi_1} = \bar{E}_1 \bar{a}_{\phi}$$

* Superposn.

- Addn. of individual Electric fields on a particular charge is the total E.

$$\bar{E}_{tot} = \bar{E}_1 + \bar{E}_2 + \dots + \bar{E}_n$$

$\rightarrow$ Example:

pt. charges q_1 & q_2 resp. located at $(4, 0, -3)$ and $(2, 0, 1)$.

$$q_2 = 4 \text{ nC.}$$

Find q_1 such that $\bar{E}$ at $(5, 0, 6)$ has no x -component.

$\rightarrow$ Solution:

$$\bullet \bar{E}_1 = \frac{q_1 k}{r^2} \hat{a}_r$$

$$= \frac{q_1 (9)(10^9)}{82} \left[\frac{\bar{a}_x + 9\bar{a}_z}{\sqrt{82}} \right]$$

$$= 0.012 (q_1) (\bar{a}_x + 9\bar{a}_z)$$

$$= 0.012 q_1 \bar{a}_x + 0.108 q_1 \bar{a}_z$$

$(10^9) \qquad \qquad \qquad (10^9)$

• Similarly,

$$\vec{E}_2 = 0.543 \vec{a}_x + 0.905 \vec{a}_y$$

$$\therefore \vec{E}_{\text{tot}} = \vec{E}_1 + \vec{E}_2$$

$$= [0.012 (10^9) q_1 + 0.543] \vec{a}_x + [0.108 (10^9) q_1 + 0.905] \vec{a}_y$$

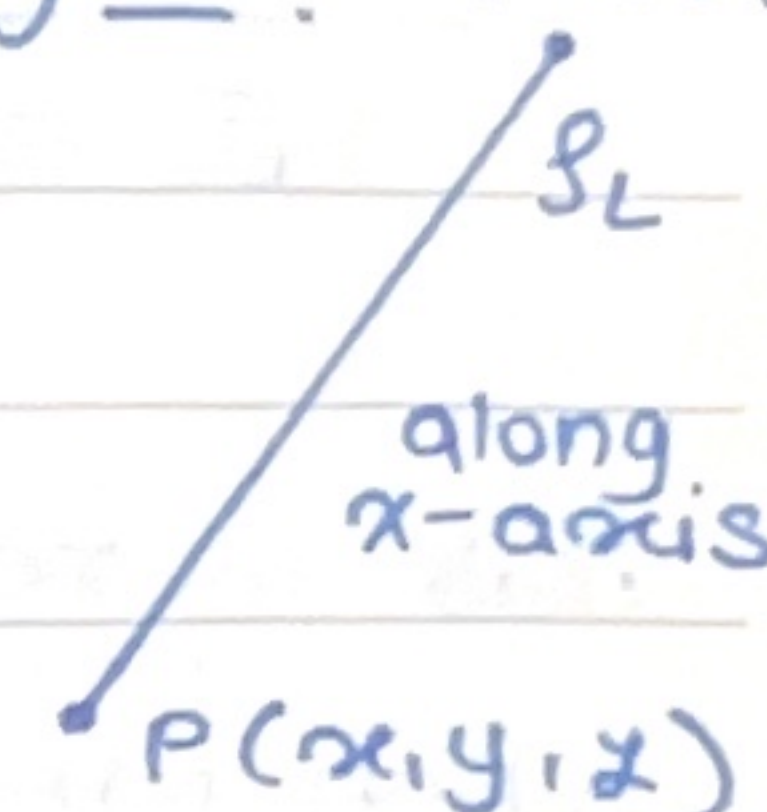
$\therefore x$ component of E_{tot} is zero

$$\therefore [0.108 (10^9) q_1 + 0.905 (10^9)] \vec{a}_y = 0 \vec{a}_y$$

$$\therefore q_1 = -8.37 \text{ nC}$$

* Electric field due to line charge: (x, y, z)

$$\vec{E} = \frac{\rho_L}{2\pi\epsilon_0 \cdot R} \hat{a}_R \quad (\text{V/m})$$



→ Example:

On the line described by $x = 2\text{m}$ and $y = -4\text{m}$, there is uniform q -distribution, $\rho_L = 20 \text{ nC/m}$, determine $\vec{E}$ at $(-2, -1, 4) \text{ m}$

→ Solution:

$\therefore$ For line q , x & y are constants & is $\parallel$ to x -axis hence, the co-ordinates of line q can be written as $(2, -4, 4)$ ie. x -coordinate is same.

$$\therefore \vec{R} = (-2-2)\vec{a}_x + (-1+4)\vec{a}_y + (4-4)\vec{a}_z$$

$$= -4\vec{a}_x + 3\vec{a}_y$$

$$\therefore \hat{a}_R = \frac{-4\vec{a}_x + 3\vec{a}_y}{5}$$

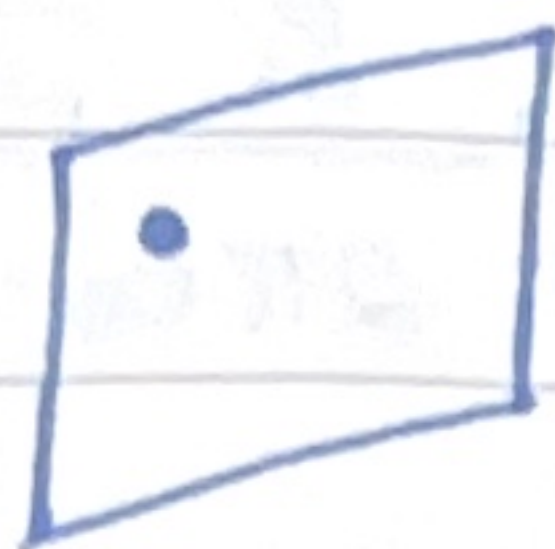
$$\frac{(-2, -1, 4) \vec{E}}{\rho_L (2, -4, 4)}$$

$$\therefore \vec{E} = \frac{20 (10^{-9})}{2\pi [8.85 \times 10^{-12}] (5)} \left[\frac{-4\vec{a}_x + 3\vec{a}_y}{5} \right]$$

$$\therefore \vec{E} = (-57.6 \vec{a}_x + 43.2 \vec{a}_y) \text{ V/m}$$

* Electric field due to infinite sheet charge:

$$\vec{E} = \frac{\rho_s}{2\epsilon_0} \vec{a}_x \quad (\text{V/m})$$



- This formula is applicable when we desire $\vec{E}$ above xy plane.
- The pt. where we desire $\vec{E}$ if below xy plane, the unit vector $\hat{r}$ changes to $-\vec{a}_x$

→ Example:

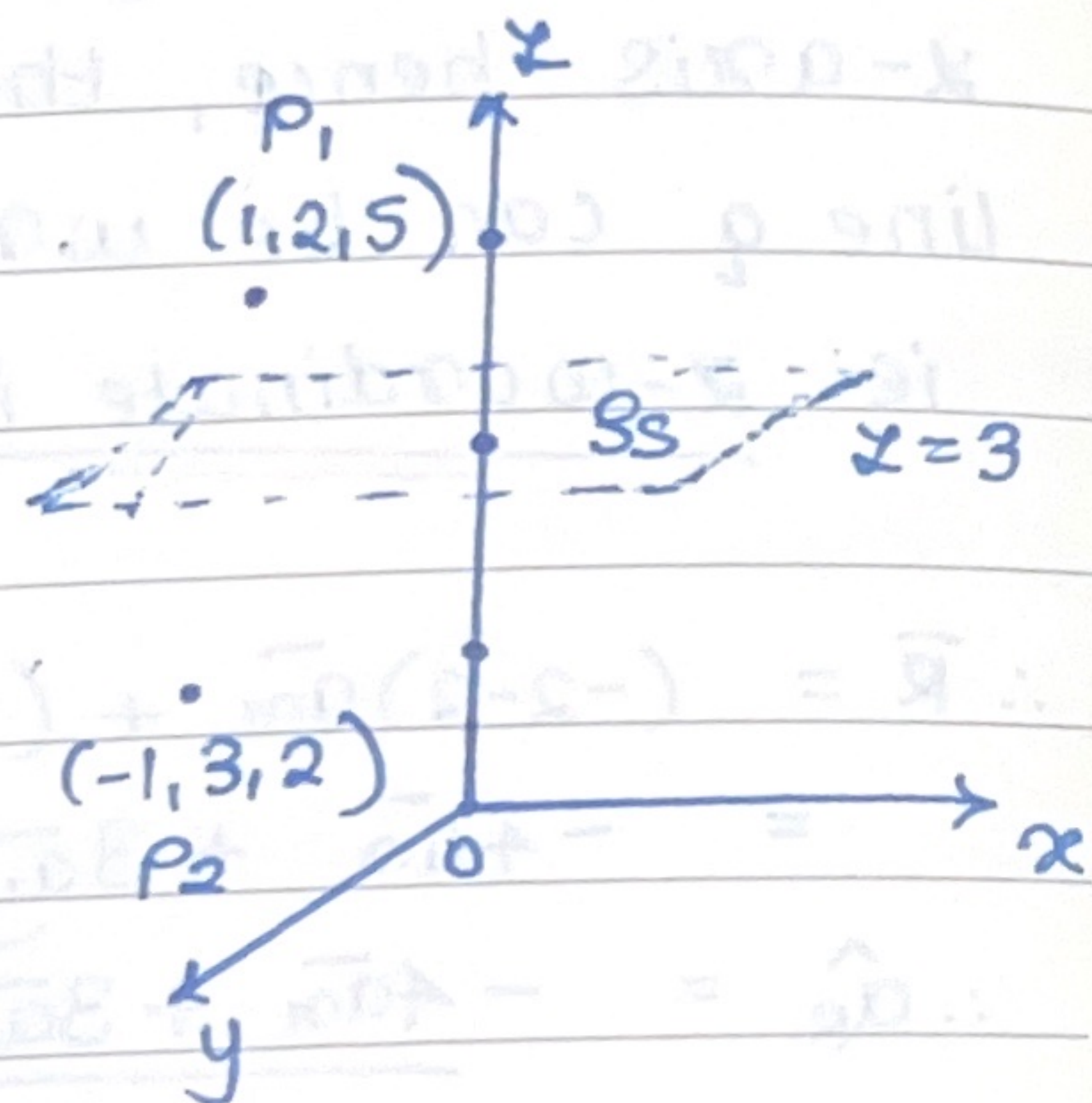
$\rho_s = (0.1) \text{ nC/m}^2$ exists on plane $x=3$.
Find $\vec{E}$ at $(1, 2, 5)$ and also at $(-1, 3, 2)$

→ Solution:

⊙ $\vec{E}$ at P_1 :

$$\therefore \vec{a}_n = \vec{a}_x$$

$$\begin{aligned} \therefore \vec{E} &= \frac{(0.1)(10^{-9})}{2(8.85)(10^{-12})} \vec{a}_x \\ &= 5.64 \vec{a}_x \quad \text{V/m} \end{aligned}$$



⊙ $\vec{E}$ at P_2 :

$\therefore$ the point lies below the infinite sheet charge ρ_s at $x=3$

$$\therefore \hat{a}_n = -\vec{a}_x$$

$$\begin{aligned} \therefore \vec{E} &= \frac{(0.1)(10^{-9})}{2(8.85)(10^{-12})} (-\vec{a}_x) \\ &= -5.64 \vec{a}_x \end{aligned}$$

Q] Find $\vec{E}$ at the origin if following charge distributions are present in free space.

- $q = 12 \text{ nC}$ at $P(2, 0, 6)$
- uniform $\rho_L = 3 \text{ nC/m}$ at $x = -2$ & $y = 3$.
- uniform $\rho_S = 0.2 \text{ nC/m}^2$ at $x = 2$.

→ Solution:

a) Due to q :

$$\hat{a}_R = \frac{-2\bar{a}_x - 6\bar{a}_y}{\sqrt{40}}$$

$$\therefore \vec{E}_0 = \frac{q}{4\pi\epsilon_0 R^2} \hat{a}_R$$

$$= \frac{(9)(10^9)(12)(10^{-9})}{40} \left[\frac{-2\bar{a}_x - 6\bar{a}_y}{\sqrt{40}} \right]$$

$$= 0.426 [-2\bar{a}_x - 6\bar{a}_y]$$

$$= -0.853 \bar{a}_x - 2.556 \bar{a}_y$$

b) Due to ρ_L :

$$\hat{a}_R = \frac{2\bar{a}_x - 3\bar{a}_y}{\sqrt{13}}$$

$$\therefore \vec{E}_L = \frac{\rho_L}{2\pi\epsilon_0 R} \hat{a}_R$$

$$= \frac{(3)(10^{-9})(18)(10^9)}{\sqrt{13}} \left[\frac{2\bar{a}_x - 3\bar{a}_y}{\sqrt{13}} \right]$$

$$= 8.3 \bar{a}_x - 12.45 \bar{a}_y$$

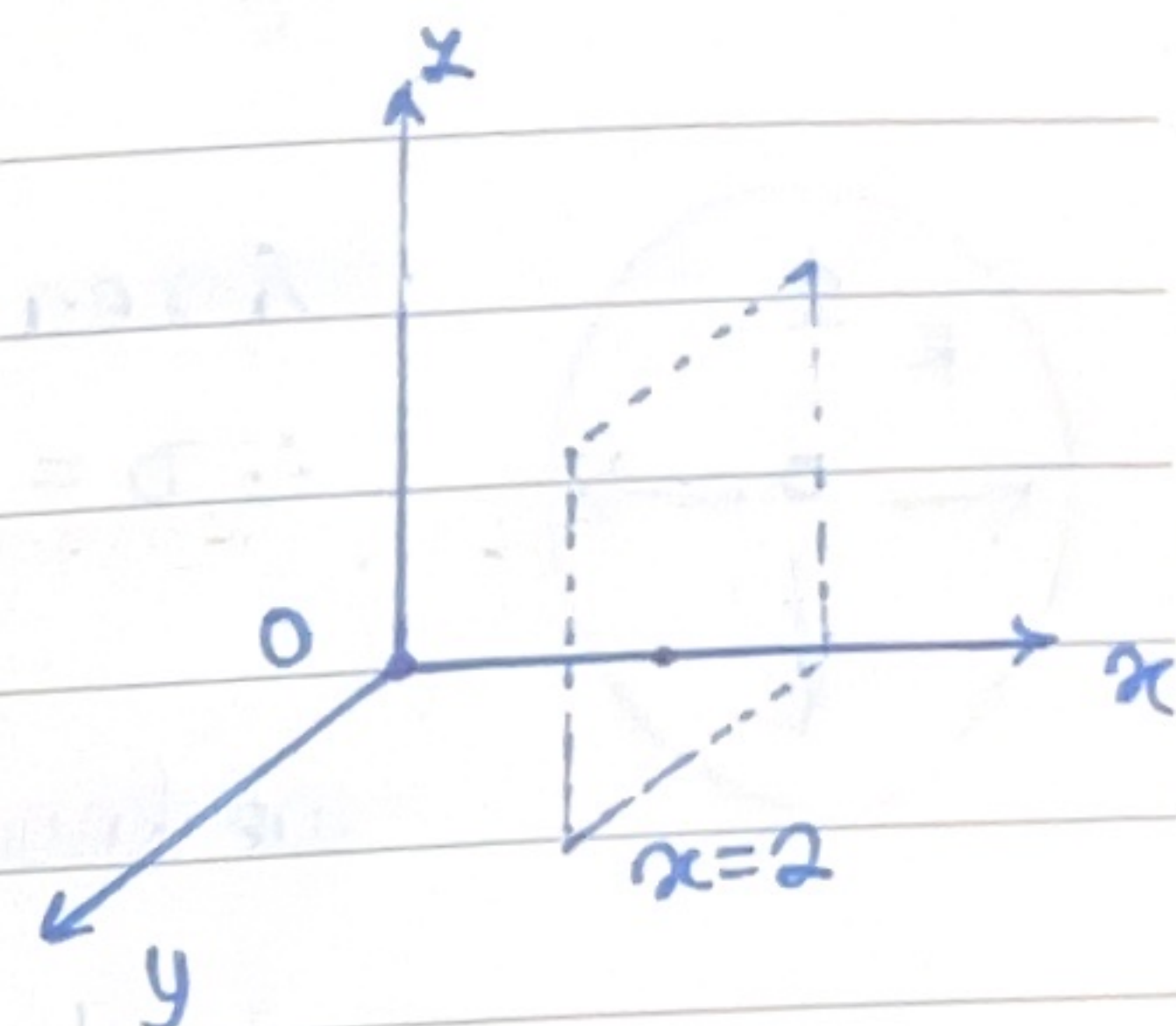
c) Due to ρ_S

$$\hat{a}_n = -\bar{a}_x$$

$$\therefore \vec{E}_S = \frac{\rho_S}{2\epsilon_0} \hat{a}_n$$

$$= \frac{(0.2)(10^{-9})}{2(8.85)(10^{-12})} (-\bar{a}_x)$$

$$= -11.2 \bar{a}_x$$



→ Thus, $\vec{E}_{\text{tot}} = \vec{E}_0 + \vec{E}_L + \vec{E}_S$

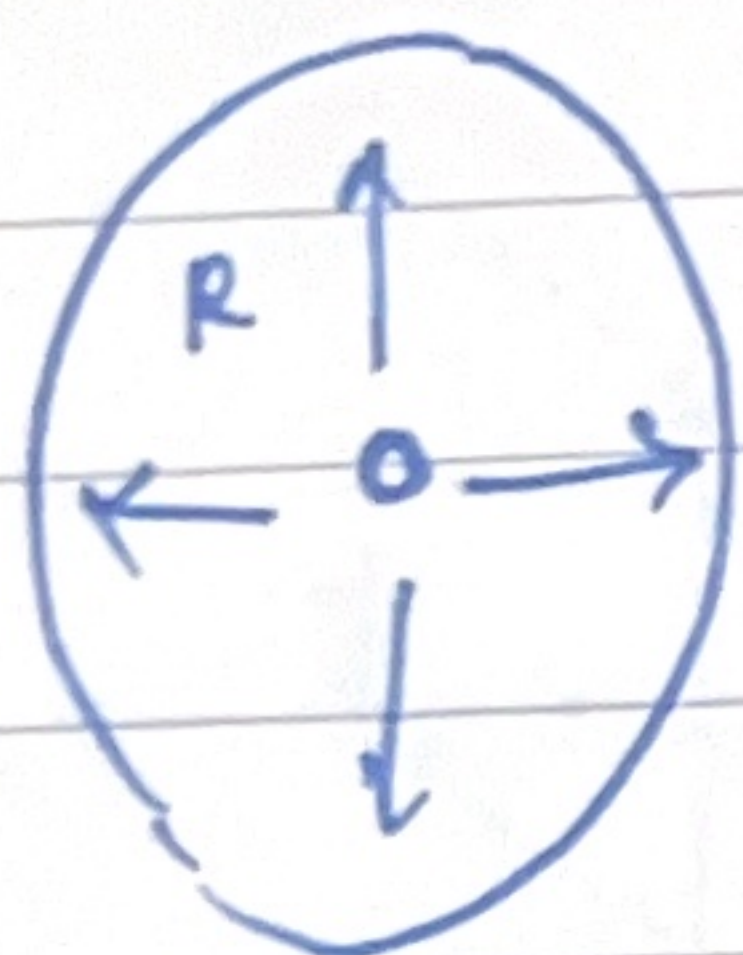
$$= (-0.853 \bar{a}_x - 2.556 \bar{a}_y) +$$

$$(8.3 \bar{a}_x - 12.45 \bar{a}_y) + (-11.2 \bar{a}_x)$$

$$= -3.85 \bar{a}_x - 12.456 \bar{a}_y - 2.56 \bar{a}_y$$

* Electric flux density :

- Total no. of field lines are called as electric flux.
- Electric flux density / Electric displacement vector field that measures amount of electric flux passing + through unit surface area is called electric flux density.



• Area of Sphere = $4\pi R^2$

$$\therefore D = \frac{Q}{A} = \frac{Q}{4\pi R^2} \quad (C/m^2)$$

we know, $E = \frac{Q}{4\pi\epsilon_0 R^2} \quad (V/m)$

thus,

$$D = \epsilon_0 \cdot E$$

a) For point charge at origin.

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_r \quad (V/m)$$

$$\therefore \vec{D} = \frac{Q}{4\pi R^2} \hat{a}_r \quad (C/m^2)$$

b) For infinite line charge :

$$\vec{E} = \frac{S_L}{2\pi\epsilon_0 R} \hat{a}_r \quad (V/m)$$

$$\therefore \vec{D} = \frac{S_L}{2\pi R} \cdot \hat{a}_r \quad (C/m^2)$$

c) For infinite surface charge :

$$\vec{E} = \frac{S_S}{2\epsilon_0} \hat{a}_n \quad (V/m)$$

$$\therefore \vec{D} = \frac{S_S}{2} \hat{a}_n \quad (C/m^2)$$

d) For volume charge density :

$$\vec{D} = \int_V \frac{S_V}{4\pi R^2} dv \hat{a}_r$$

→ Example:

A pt. charge of $6\mu\text{C}$ is located at o ,
a uniform $\rho_L = 180\text{ nC/m}$ lies along
 x -axis and $\rho_S = 25\text{ nC/m}^2$ lies in
a $z=0$ plane.

Find $\vec{D}$ at $(0,0,4)$, calculate total
electric flux leaving the surface
of sphere having $R=4\text{ m}$ centered at o .

→ Solution:

(i) a) $\vec{D}$ due to pt. charge:

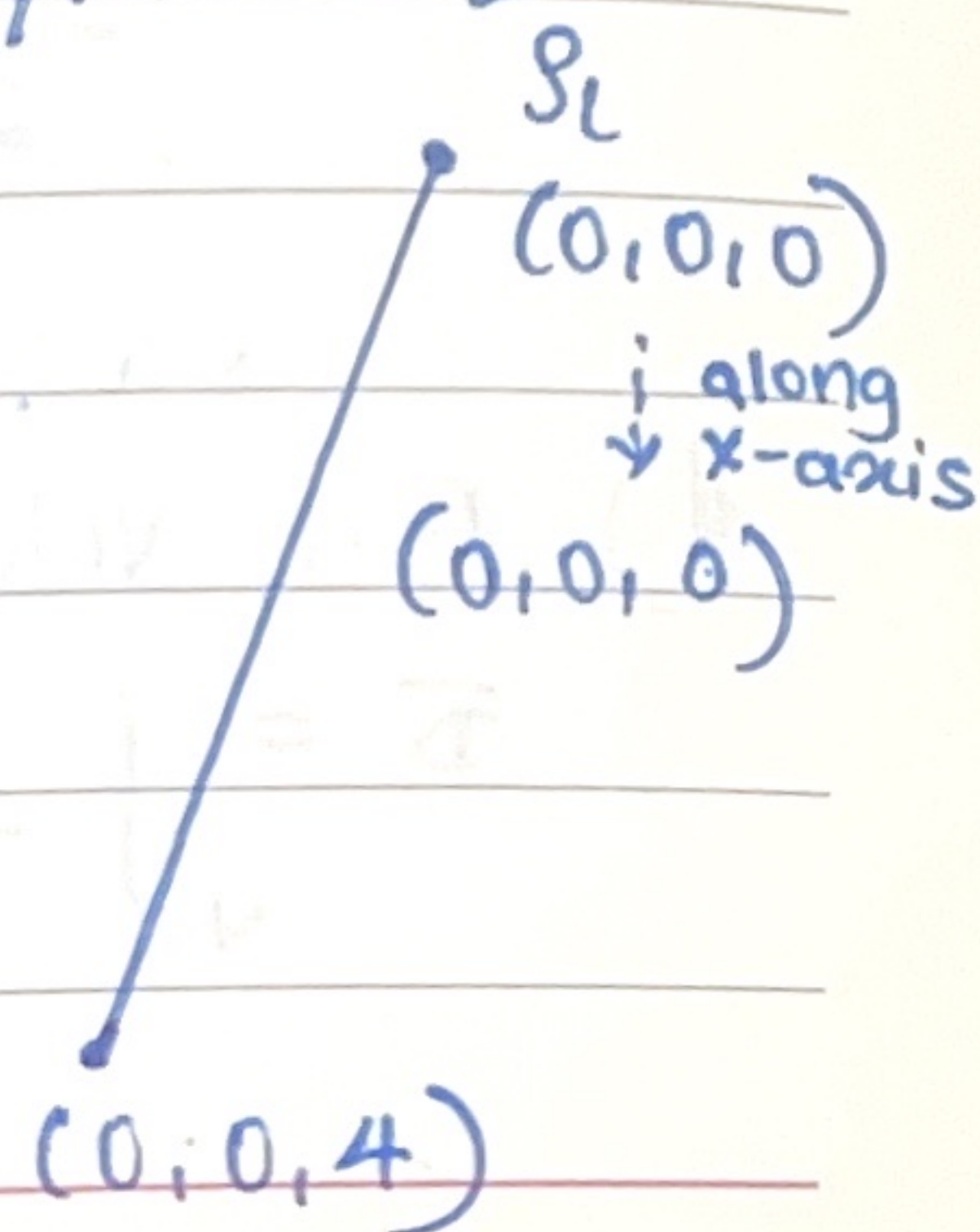
$$\hat{a}_r = \frac{4\hat{a}_y}{4} = \hat{a}_y$$

$$\therefore \vec{D}_0 = \frac{q}{4\pi R^2} \hat{a}_r \text{ (nC/m}^2\text{)}$$

$$= \frac{(6)(10^{-6})}{4\pi(4)^2} \hat{a}_y = 29.8\text{ nC/m}^2 \hat{a}_y$$

b) $\vec{D}$ due to line charge:

$$\hat{a}_r = \frac{4\hat{a}_y}{4} = \hat{a}_y$$



$$\therefore \vec{D}_L = \frac{(180)(10^{-9})}{2\pi(4)} \hat{a}_y$$

$$= 7.16 \hat{a}_y \text{ nC/m}^2$$

c) $\vec{D}$ due to surface charge:

$$\vec{D}_S = \frac{(25)(10^{-9})}{2} \hat{a}_y$$

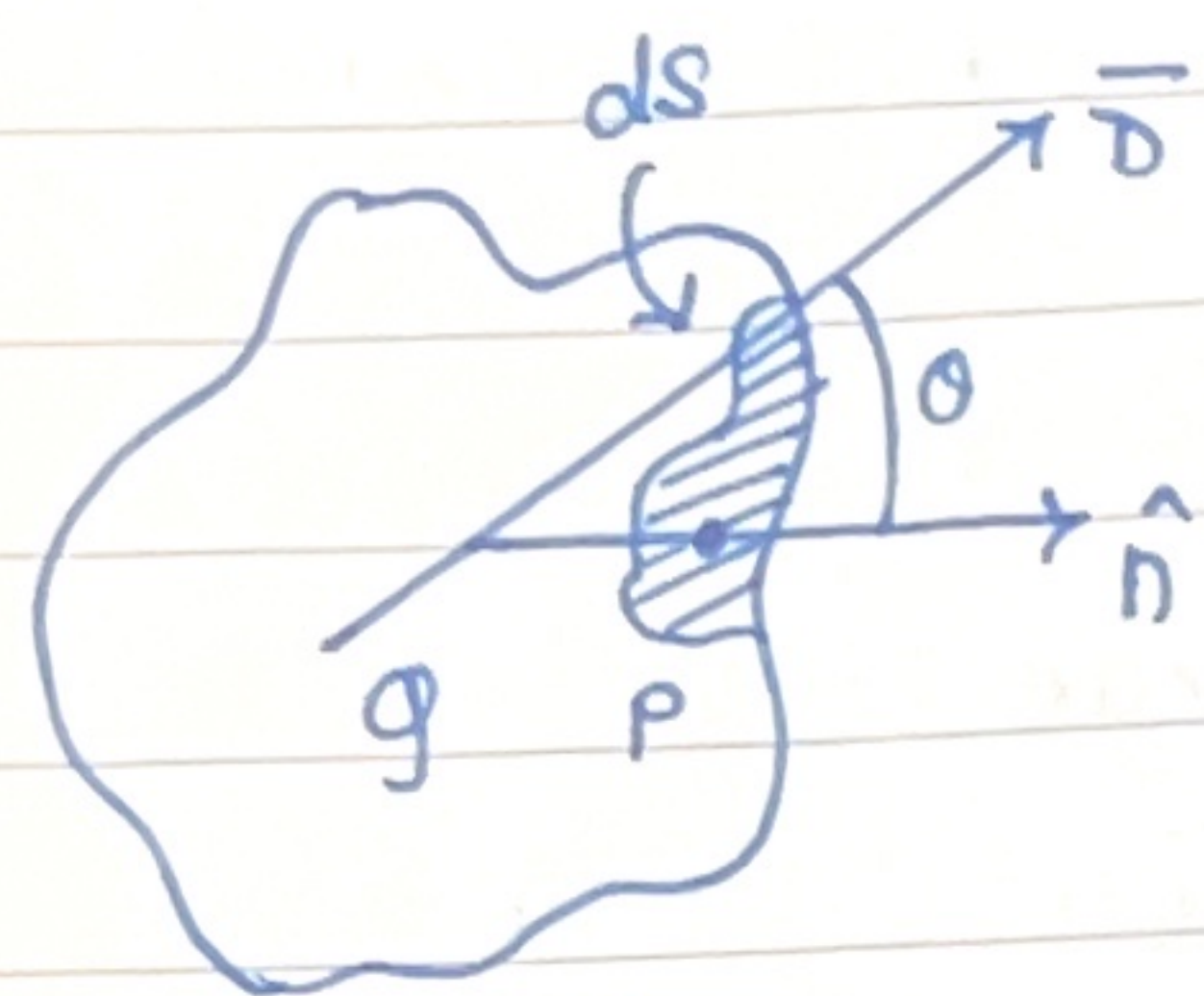
$$= 12.5 \hat{a}_y \text{ nC/m}^2$$

→ Thus,

$$\begin{aligned} \vec{D}_{\text{tot}} &= \vec{D}_0 + \vec{D}_L + \vec{D}_S \\ &= [29.8 \hat{a}_y + 7.16 \hat{a}_y + 12.5 \hat{a}_y] \text{ nC/m}^2 \\ &= 49.46 \hat{a}_y \text{ nC/m}^2 \end{aligned}$$

(ii) The total Q_{enc} .

$$\begin{aligned} &= q_0 + (\text{diam.}) \rho_L + (\text{Area}) \rho_S \\ &= 6(10^{-6}) + 8(180)(10^{-9}) + \pi(4)^2(25)(10^{-9}) \\ &= 8.696 \mu\text{C} \end{aligned}$$

* Gauss' law :

- It states that $\vec{D}$ passing through any closed surface = Total charge enc. by that surface.

ie. $\oint \vec{D} \cdot d\vec{s} = q_{enc.}$

→ Mathematical expression :

- Here, +ve charge 'q' is enclosed by closed surface of any shape. At pt. P, consider differential element of surface 'ds' and let $\vec{D}$ makes an angle 'θ' with ds, where.

$$d\vec{s} = ds \cdot \vec{a}_n$$

here, $\vec{a}_n$ = vector along $\hat{n}$

- The flux crossing $d\vec{s}$ is product of Normal component of $\vec{D}$ & $d\vec{s}$

$$\therefore \oint \vec{D} \cdot d\vec{s} = q_{enc.} = \psi$$

→ proof :

- Consider total flux passing through Sphere :

$$\psi = \oint \vec{D} \cdot d\vec{s}$$

- ∵ pt. charge is situated at centre, flux density is :

$$\vec{D} = \frac{q}{4\pi r^2} \cdot \vec{a}_r$$

- The diff. area on spherical surface is $ds = r^2 \sin \theta \, d\theta \cdot d\phi \cdot \vec{a}_r$

$$\text{• Now, } \psi = \oint \frac{q}{4\pi r^2} \vec{a}_r [r^2 \sin \theta \, d\theta \cdot d\phi] \vec{a}_r$$

$$\therefore \psi = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \frac{q}{4\pi r^2} [r^2 \sin \theta \, d\theta \, d\phi]$$

$$= \frac{q}{4\pi} \int_0^{2\pi} \int_0^{\pi} \sin \theta \cdot d\theta \cdot d\phi$$

$$= \frac{-q}{4\pi} \int_0^{2\pi} [-1-1] d\phi = \frac{q}{2\pi} [2\pi - 0] = q$$

$$\therefore \oint \vec{D} \cdot d\vec{s} = q \quad \rightarrow \text{HENCE PROVED}$$

Q] A circular disc of $r = 1\text{m}$ with charge density $(\rho_s) = 2 \sin \phi \text{ uC/m}^2$ is enclosed by surface 's'.
How much flux crosses 's'?

→ Solution:

$$\psi = q_{\text{enc.}}$$

$$= \oint \rho_s \cdot ds$$

$$= \int_{r=0}^1 \int_{\phi=0}^{2\pi} 2 \sin \phi \cdot r \cdot dr \cdot d\phi$$

$$= 2 \left[\frac{r^2}{2} \right]_0^1 \left[-\cos \phi \right]_0^{2\pi}$$

$$= [-1 + 1]$$

$$= 0$$

Q] A sphere of $r = a \text{ m}$ is centered at $O(0,0,0)$ and has uniform $\rho_s \text{ C/m}^2$, using Gauss' law, determine $\vec{E}$ in all regions the medium is free space

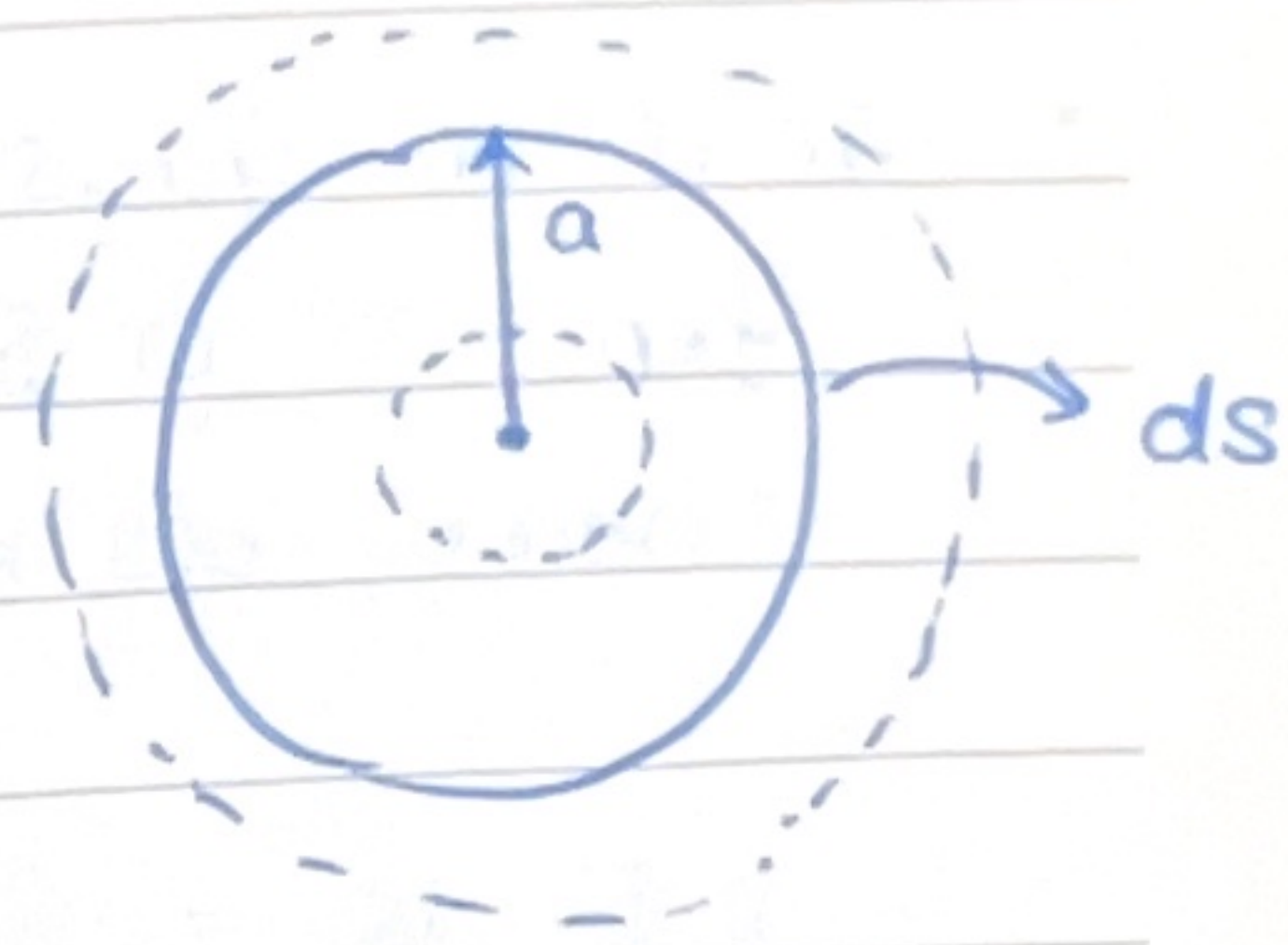
→ Solution:

$$\cdot \quad q = \oint \vec{D} \cdot d\vec{s}$$

$$\text{and } \vec{D} = \epsilon_0 \vec{E}$$

There are 2 region:

a) $r < a$ b) $r > a$



a) when $r < a$:

∴ The q is present on sphere of radius $= a$, the inner sphere doesn't enclose any charge

$$\therefore q_{\text{enc}} = 0$$

$$\therefore \vec{D} = 0 \Rightarrow \vec{E} = 0$$

b) when $r > a$:

• It'll enclose the given sphere

$$\therefore q_{\text{enc}} = \int \rho_s \cdot ds$$

$$\therefore q_{\text{enc}} = \rho_s (s)$$

$$= \rho_s (4\pi a^2)$$

- Accd. to Gauss' law:

$$Q_{enc} = \oint \vec{D} \cdot d\vec{s}$$

$$\text{here, } d\vec{s} = r^2 \sin \theta \cdot d\theta \cdot d\phi$$

$$\therefore \oint \vec{D} \cdot d\vec{s} = \rho_s (4\pi a^2)$$

$$\therefore \oint \vec{D} \cdot d\theta \cdot d\phi = \rho_s (4\pi a^2)$$

$$\therefore \int_0^\pi d\theta \int_0^{2\pi} \vec{D} \cdot \sin \theta \cdot d\phi = \frac{\rho_s (4\pi a^2)}{r^2}$$

$$\therefore \vec{D} [2\pi - 0] = \frac{\rho_s (4\pi a^2)}{r^2}$$

$$\therefore \vec{D} = \frac{\rho_s a^2}{r^2}$$

- Now,

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\therefore \vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{\rho_s a^2}{\epsilon_0 r^2}$$

- Example:

3 concentric spherical sources of radii $r = 3, 4, 5$ m have uniform $\rho_s = 8, -12, \rho_s$ nC/m²

i) what will be ρ_s so as to make $D = 0$ for $r > 5$

ii) ~~F~~

→ Solution:

- We know, $\vec{D} = \rho_s \cdot \frac{a^2}{r^2}$

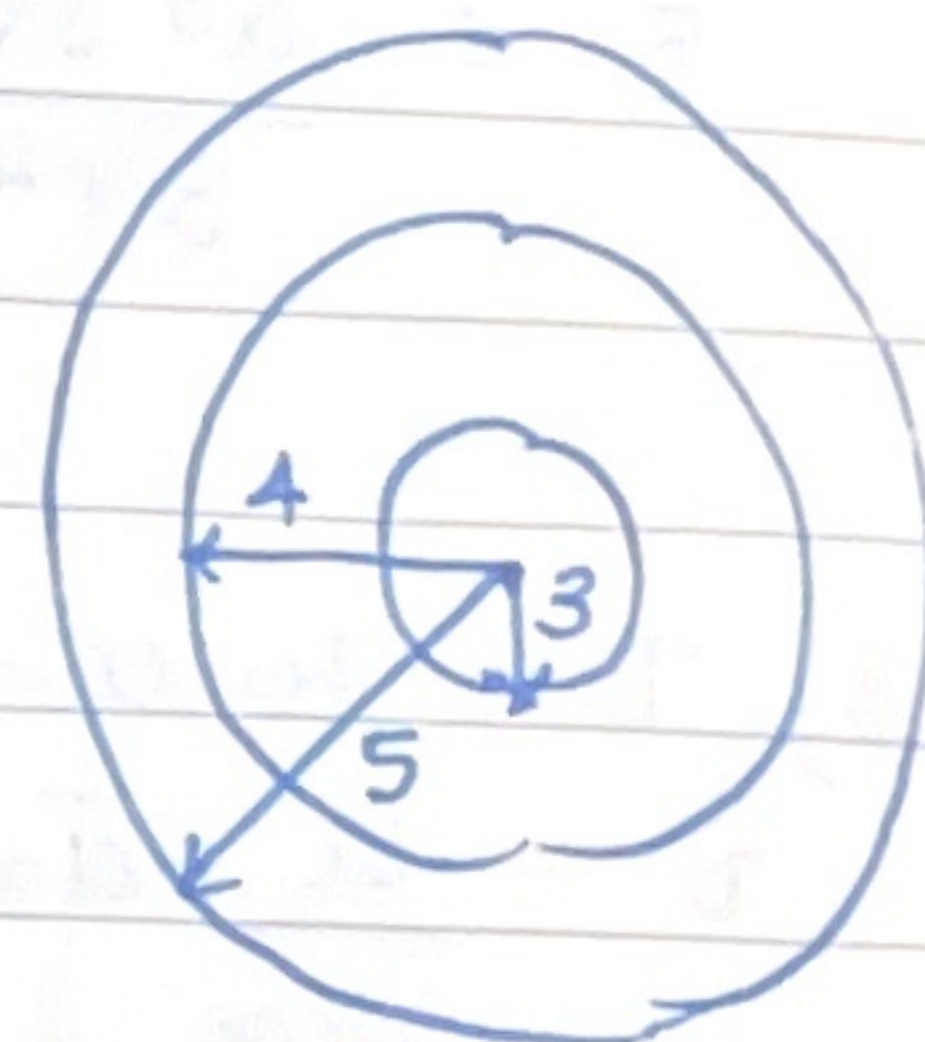
$$\therefore D_1 = (8)(10^{-9}) \frac{(3)^2}{r^2} \vec{a}_r$$

$$\therefore D_2 = (-12)(10^{-9}) \frac{(4)^2}{r^2} \vec{a}_r$$

$$\therefore D_3 = \rho_s (10^{-9}) \frac{(5)^2}{r^2} \vec{a}_r$$

- Now, $D_{tot} = D_1 + D_2 + D_3 = 0$

$$\therefore \rho_s = 4.8 \text{ nC/m}^2$$



$$r_1 = 3 \text{ m}, \rho_{s1} = 8 \text{ nC/m}^2$$

$$r_2 = 4 \text{ m}, \rho_{s2} = -12$$

$$r_3 = 5 \text{ m}, \rho_{s3} = \rho_s$$

General Formulas

→ Due to uniformly charged Sphere :

$$1) \bar{D} = \rho_s \cdot \frac{a^2}{r^2} \bar{a}_r$$

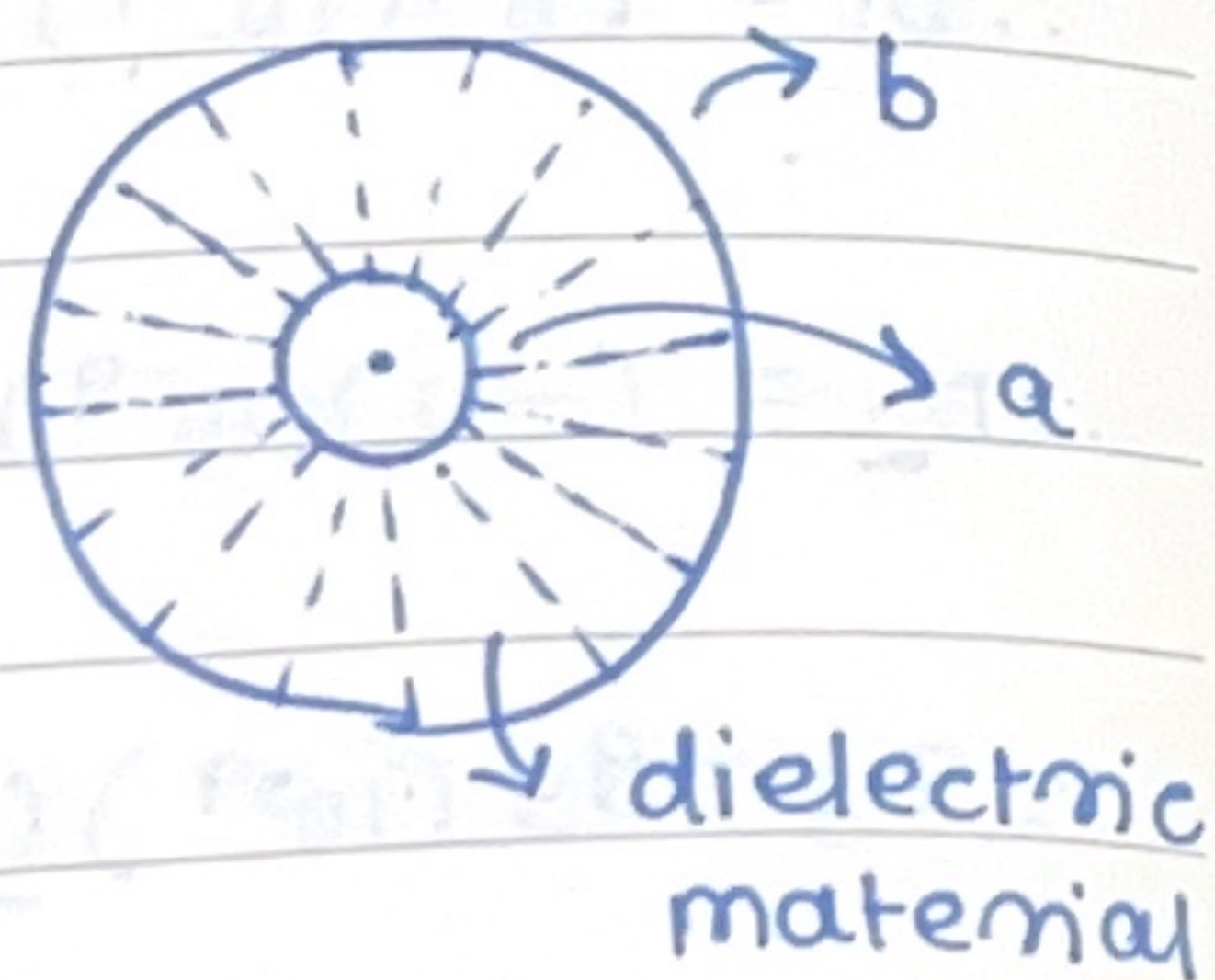
$$2) \bar{D} = \frac{\rho_v(r)}{3} \bar{a}_r ; (r < a)$$

$$\bar{D} = \frac{a^3 \rho_v}{3r^2} \bar{a}_r ; (r > a)$$

3) Due to co-axial cable

$$\bar{D} = \frac{\rho_l}{2\pi r} \bar{a}_r ; (a < r < b)$$

$$= 0 ; (r > b \text{ \& } r < a)$$



→ Example :

A uniform vol. charge density of $80 \mu\text{C/m}^3$ is present throughout the region $(8 \text{ mm} < r < 10 \text{ mm})$

i) Find total charge inside spherical surface, $r = 10 \text{ mm}$

ii) Find $\bar{D}_r$ at $r = 10 \text{ mm}$

iii) Find $\bar{D}_r$ at $r = 20 \text{ mm}$

→ Solution :

$$i) q_{enc} = \int_V \rho_v \cdot dv$$

$$= \int_0^{2\pi} \int_0^{\pi} \int_{r=8\text{mm}}^{10\text{mm}} (80)(10^{-6}) r^2 \sin \theta \cdot dr \cdot d\theta \cdot d\phi$$

$$= 163.53 \text{ pC}$$

$$ii) \bar{D} = \frac{q}{A} = \frac{163.53 (10^{-12})}{4\pi [10\text{mm}]^2} = 13 \mu\text{C/m}^2$$

$$iii) \bar{D} = \frac{q}{A} = \frac{163.53 (10^{-12})}{4\pi [2\text{mm}]^2} = 32.5 \text{ nC/m}^2$$

* Divergence Theorem:

- It states that integral of normal component of vector function over a closed surface equal to integral of div. of that vector, throughout the volume V , enclosed by surface S

$$\text{ie. } \oint \vec{D} \cdot d\vec{s} = \int_V (\nabla \cdot \vec{D}) dV$$

here, $\nabla \cdot \vec{D} = \rho_v$ (pt. form of Gauss' law)

i] for cartesian system:

$$\nabla \cdot \vec{D} = \frac{\partial}{\partial x} (D_x) + \frac{\partial}{\partial y} (D_y) + \frac{\partial}{\partial z} (D_z)$$

ii] for cylindrical system:

$$\nabla \cdot \vec{D} = \frac{1}{r} \frac{\partial}{\partial r} (r D_r) + \frac{1}{r} \frac{\partial}{\partial \phi} (D_\phi) + \frac{\partial}{\partial z} (D_z)$$

iii] for Spherical system:

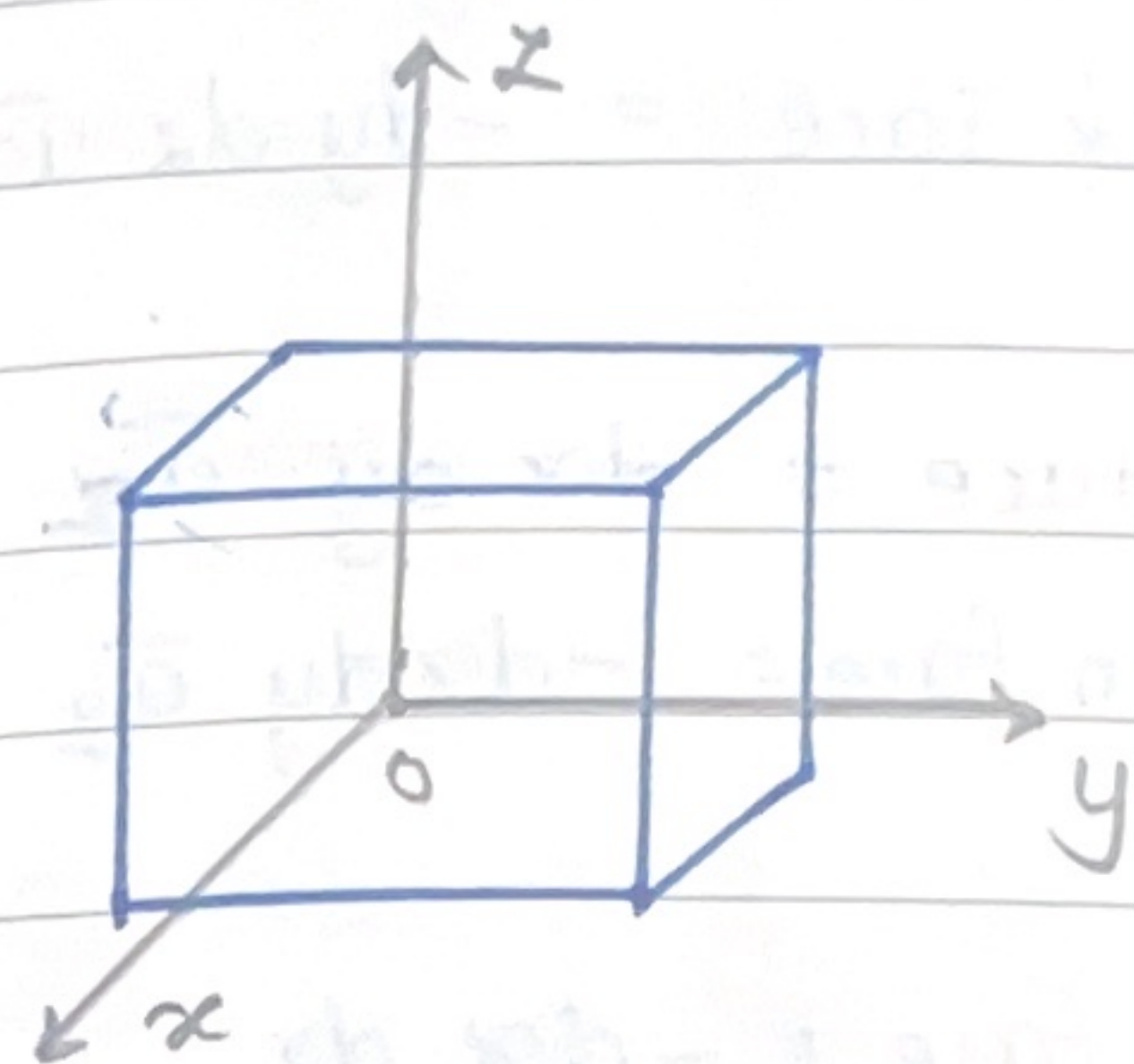
$$\nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cdot D_\theta) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} (D_\phi)$$

→ Example:

$$\vec{D} = \left[\frac{10}{3} \right] x^3 \vec{a}_x \text{ (C/m}^2\text{)}$$

evaluate both sides of div. thm. for the volume of a cube of 2m edge, centered at the origin at edges || to axis

→ Solution:



$$(A) \cdot \oint \vec{D} \cdot d\vec{s} = \int_V \nabla \cdot \vec{D} \cdot dV, \text{ acc. to div thm.}$$

Given structure is a cube which has 6 faces: F, B, R, L, T, Bot. ; $\therefore$ total 6 ds.

$$\therefore \int_S \vec{D} \cdot d\vec{s} = \int_F \vec{D} \cdot d\vec{s} + \int_B \vec{D} \cdot d\vec{s} + \int_R \vec{D} \cdot d\vec{s} + \int_L \vec{D} \cdot d\vec{s} + \int_T \vec{D} \cdot d\vec{s} + \int_{Bot} \vec{D} \cdot d\vec{s}$$

NOTES

Date

$\therefore$ The cube is centered at origin, hence,
the front face is at $x=1$ & back face: $x=-1$
Similarly, left: $y=-1$ & right: $y=1$
top: $z=1$ & bottom: $z=-1$

$\rightarrow$ For these Surfaces:

a) ds for front face = $dy \, dz \, \bar{a}_x$ ($x=1$)
ds for back face = $-dy \, dz \, \bar{a}_x$ ($x=-1$)

b) $\rightarrow$ top face = $dx \, dy \, \bar{a}_z$ ($z=1$)
 $\rightarrow$ bottom face = $-dx \, dy \, \bar{a}_z$ ($z=-1$)

c) $\rightarrow$ left face = $-dx \, dz$ ($y=-1$)
 $\rightarrow$ right face = $dx \, dz$ ($y=1$)

$\rightarrow \therefore \vec{D} = \left(\frac{10}{3}\right) x^3 \bar{a}_x$ has only $\bar{a}_x$

component.

$\therefore D_{top} = D_{bot} = D_R = D_L = 0$

NOTES

Date

$$\therefore \oint \vec{D} \cdot d\vec{s} = \int_{-1}^1 \int_{-1}^1 \left[\frac{10}{3} x^3 \bar{a}_x \right] dy \, dz \, \bar{a}_x$$

$$+ \int_{-1}^1 \int_{-1}^1 \left[\frac{10}{3} x^3 \bar{a}_x \right] dy \, dz (-\bar{a}_x)$$

$$= \frac{10}{3} (1)^3 [2][2] + \left[-\frac{10}{3} (-1)^3 [2][2] \right]$$

$$= \frac{40}{3} - \left[-\frac{40}{3} \right] = \frac{80}{3}$$

(B) Now,

$$\nabla \cdot \vec{D} = \frac{\partial}{\partial x} D_x + \frac{\partial}{\partial y} D_y + \frac{\partial}{\partial z} D_z$$

$$= \frac{10}{3} (3x^2) = 10x^2$$

$$\therefore \int_V \nabla \cdot \vec{D} \, dv = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 (10x^2) dx \, dy \, dz$$

$$= 40 \left[\frac{1}{3} + \frac{1}{3} \right] = \frac{80}{3}$$

* Work done:

- $W = F \times \text{disp.}$

ie. $W = \int dw$

Total w. req. to move the q , the finite distance betn. 2 pts. from initial to final posn. is obtained by

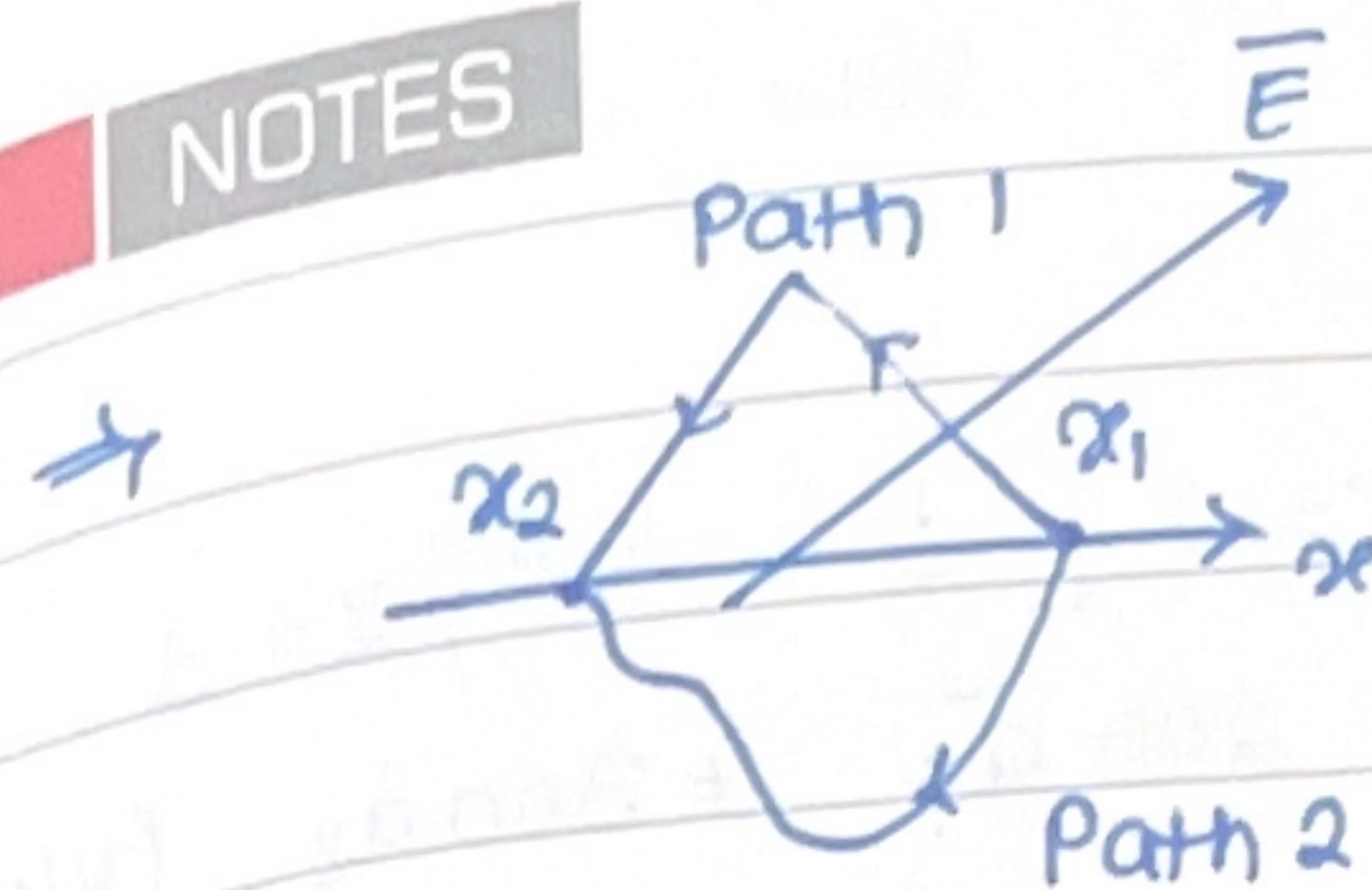
integrating dw
ie. $W = -q \int_{\text{ini}}^{\text{fin.}} \vec{E} \cdot d\vec{l}$

- $d\vec{l}$ varies w.r.t. cartesian, cylindrical, spherical

$$d\vec{l} = dx \vec{a}_x + dy \vec{a}_y + dz \vec{a}_z$$

$$d\vec{l} = dr \vec{a}_r + r d\phi \vec{a}_\phi + dz \vec{a}_z$$

$$d\vec{l} = dr \vec{a}_r + r d\theta \vec{a}_\theta + r \sin \theta \cdot d\phi \cdot \vec{a}_\phi$$



- A pt. charge q can follow 2 diff. paths but the w. done will remain same

- $\therefore W = q(V_A - V_B)$

V_A, V_B : potentials at pts A, B resp.

- For closed loop: $W = 0$

$$\therefore W = -q \int_{\text{ini}}^{\text{fin.}} \vec{E} \cdot d\vec{l} = 0$$

$$\therefore \int_{\text{ini}}^{\text{fin.}} \vec{E} \cdot d\vec{l} = 0$$

- By Stoke's thm.:

$$\oint_S \vec{E} \cdot d\vec{l} = \int_S (\nabla \times \vec{E}) \cdot d\vec{s}$$

$$\therefore \nabla \times \vec{E} = 0 \quad (\text{pt. form of Maxwell's eqn.})$$

→ Example :

The value of $\vec{E}$ at $r=2$, $\phi = 40^\circ$, $z=3$ is given as $\vec{E} = 100\vec{a}_r - 200\vec{a}_\phi + 300\vec{a}_z$ (V/m).

Determine incremental w (ie. dw) req. to move $20 \mu\text{C}$ charge at a distance of $6(10^{-6})\text{m}$ in dirn. of

- i) $\vec{a}_r$ iii) $\vec{a}_z$ v) $\vec{Q} = 2\vec{a}_x - 3\vec{a}_y + 4\vec{a}_z$
 ii) $\vec{a}_\phi$ iv) $\vec{E}$

→ Solution :

• Given system is cylindrical.

$$\therefore d\vec{l} = dr \cdot \vec{a}_r + r d\phi \vec{a}_\phi + dz \vec{a}_z$$

i) In dirn. of $\vec{a}_r$:

$$d\vec{l} = dr \cdot \vec{a}_r$$

$$= 6(10^{-6}) \vec{a}_r$$

$$\therefore dw = -q \vec{E} \cdot d\vec{l}$$

$$= -(20)(10^{-6}) [100\vec{a}_r - 200\vec{a}_\phi + 300\vec{a}_z] 6(10^{-6}) \vec{a}_r$$

$$= -12 \text{ nJ}$$

ii) In dirn. of $\vec{a}_\phi$:

$$d\vec{l} = r d\phi \vec{a}_\phi$$

$$= (2)(6)(10^{-6}) \vec{a}_\phi$$

$$\therefore dw = -(20)(10^{-6}) [100\vec{a}_r - 200\vec{a}_\phi + 300\vec{a}_z]$$

$$(12)(10^{-6}) \vec{a}_\phi$$

$$= 48 \text{ nJ}$$

iii) In dirn. of $\vec{a}_z$:

$$d\vec{l} = dz \cdot \vec{a}_z$$

$$= 6(10^{-6}) \vec{a}_z$$

$$\therefore dw = -(20)(10^{-6}) [100\vec{a}_r - 200\vec{a}_\phi + 300\vec{a}_z]$$

$$(6)(10^{-6}) \vec{a}_z$$

$$= -36 \text{ nJ}$$

iv) In dirn. of $\vec{E}$

$$d\vec{l} = d\vec{l} \cdot \vec{a}_E$$

$$\text{here, } \vec{a}_E = \frac{100\vec{a}_r - 200\vec{a}_\phi + 300\vec{a}_z}{374.16}$$

$$\therefore dW = -(20)(10^{-6}) \left[100 \bar{a}_x - 200 \bar{a}_\phi + 300 \bar{a}_z \right] \\ (6)(10^{-6}) \left[\frac{100 \bar{a}_x - 200 \bar{a}_\phi + 300 \bar{a}_z}{344.16} \right]$$

$$= -36 \text{ nJ}$$

v) In dirn. of $\bar{q} = 2\bar{a}_x - 3\bar{a}_y + 4\bar{a}_z$

Now,

$$\bullet Q_x = q(\bar{a}_x)$$

$$= (2\bar{a}_x - 3\bar{a}_y + 4\bar{a}_z) (\bar{a}_x)$$

$$= 2 \cos \phi - 3 \sin \phi + 4(0)$$

$$= -0.4$$

$$\bullet Q_\phi = q(\bar{a}_\phi)$$

$$= (2\bar{a}_x - 3\bar{a}_y + 4\bar{a}_z) (\bar{a}_\phi)$$

$$= -2 \sin \phi - 3 \cos \phi + 4(0)$$

$$= -3.8$$

$$\bullet Q_z = q(\bar{a}_z)$$

$$= (2\bar{a}_x - 3\bar{a}_y + 4\bar{a}_z) (\bar{a}_z)$$

$$= 4$$

$$\therefore \bar{q} = -0.4\bar{a}_x - 3.8\bar{a}_\phi + 4\bar{a}_z$$

$$\therefore d\bar{l} = dl \cdot \bar{a}_q$$

$$= 6(10^{-6}) \left[\frac{-0.4\bar{a}_x - 3.8\bar{a}_\phi + 4\bar{a}_z}{5.53} \right]$$

thus,

$$dW = -(20)(10^{-6}) \left[100 \bar{a}_x - 200 \bar{a}_\phi + 300 \bar{a}_z \right] \\ (6)(10^{-6}) \left[\frac{-0.4\bar{a}_x - 3.8\bar{a}_\phi + 4\bar{a}_z}{5.53} \right]$$

$$= \frac{-(21.7)}{(10^{-12})} \left[-40 + 760 + 1200 \right]$$

$$= -41.6 \text{ nJ}$$